Locally Honest Coverage Forces Nonparametric Inflation: a Sharpness Lower Bound for Prediction Sets

Peter Cotton
peter.cotton@microprediction.com

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Abstract

Marginal conformal coverage costs no sharpness and admits oracle-rate bands. A guarantee at a fixed test point is different. We show that any prediction set with honest coverage at an interior point, valid over a Hölder class of residual laws, exceeds the oracle width by at least the local minimax modulus $(n\,g(x_0))^{-s/(2s+d)}$. The cause is Le Cam indistinguishability, not exchangeability. A local increase in spread of that size is invisible in the sample yet moves the true quantile, so an honest set must insure against it. A plug-in estimate is sharper because it does not. Under unknown smoothness, honesty cannot adapt.

1 Two guarantees, one honest

Fix a model, freeze it, and let $R = A(X, Y)$ be its scalar nonconformity score with conditional law $\mathcal{L}(R \mid X = x)$. Ordinary conformal coverage is

$$\mathbb{P}\big\{R_{n+1} \in C_n(X_{n+1})\big\} \;\geq\; 1-\alpha, \tag{1}$$

the probability averaging over the test point X_{n+1} and the calibration sample D_n. Under (1) no impossibility holds: Lei and Wasserman (2014) construct bands with exact finite-sample marginal validity that converge to an oracle band at the minimax rate, and localized conformal methods retain marginal validity while weighting by proximity. A theorem about the cost of a guarantee must therefore concern a guarantee that is local. We take the honest one.

Definition 1 (Honest local coverage)**.** Fix an interior point x_0 and $0 \leq \eta < \frac{1}{2}$. A one-sided prediction set $C_n(x_0) = [0, T_n(x_0)]$ for a nonnegative score is *honestly locally valid* over a class Θ if

$$\inf_{\theta\in\Theta} \mathbb{P}_\theta\Big\{F_{\theta(x_0)}\big(T_n(x_0)\big) \geq 1-\alpha\Big\} \;\geq\; 1-\eta, \tag{2}$$

where $F_{\theta(x_0)}$, assumed continuous and strictly increasing, is the true conditional law of the score at x_0, so the event equals $\{T_n(x_0) \geq q_\alpha(\theta(x_0))\}$, and the probability is over the calibration sample. In words: with probability at least $1-\eta$ over the data, the realized set has conditional coverage $1-\alpha$ at x_0, whatever law in Θ generated the data. Taking $\eta = 0$ is literal per-sample conditional coverage.[1]

[1] *Honest* is the standard term for a coverage guarantee that holds uniformly over the class at finite n, the $\inf_\theta$ sitting inside the probability, as opposed to a pointwise or asymptotic guarantee that fixes θ first (Li, 1989; Low, 1997; Cai and Low, 2004). The qualifier *local* is ours: we impose it at a fixed interior point x_0 rather than over an interval.

Sharpness is measured against the oracle threshold $q_\alpha(u) = F_u^{-1}(1-\alpha)$ through the excess

$$S_\theta(T_n) \;=\; \mathbb{E}_\theta\Big[\big(T_n(x_0) - q_\alpha(\theta(x_0))\big)_+\Big], \tag{3}$$

the expected inflation of the threshold above the oracle. The question is how small (3) can be made subject to (2).

2 Assumptions

The upper Hölder bound alone is not enough for a lower bound; the class must contain alternatives that are indistinguishable near x_0. Three standing assumptions supply this without asking us to estimate an arbitrary law.

Assumption 1 (Local design regularity)**.** The covariate density g satisfies $0 < g_- \le g(x) \le g_+ < \infty$ on a fixed neighbourhood of $x_0 \in (0,1)^d$.

Assumption 2 (Regular residual family)**.** The conditional score law is $R \,|\, X = x \sim F_{\theta(x)}$ for a one-parameter family $\{F_u : u \in [\underline{u}, \overline{u}]\}$ that is, locally in u, quadratically regular in information and Lipschitz-regular in quantile:

$$\mathrm{KL}\big(F_u, F_v\big) \le K\,(u-v)^2, \qquad c_q\,|u-v| \le \big|q_\alpha(u) - q_\alpha(v)\big| \le C_q\,|u-v|.$$

A truncated exponential family with parameter in a compact interval is a concrete instance. The lower bound below uses only the quadratic information bound and the lower quantile-sensitivity constant c_q; the upper constant C_q and the design lower bound g_- of Assumption 1 enter only the attainability statement of Section 4.

Assumption 3 (Hölder smoothness with local richness)**.** The parameter field lies in a Hölder ball, $\theta \in \mathcal{H}^s(L)$, and the class contains localized perturbations of a constant baseline: for a fixed compactly supported bump ψ with $\psi(0) \neq 0$ and any $h > 0$,

$$\theta_1(x) = \theta_0(x) + \delta\,\psi\Big(\frac{x - x_0}{h}\Big) \in \mathcal{H}^s(L), \qquad \delta \asymp L\,h^s.$$

Assumption 3 is the operative one. It says the class holds a pair that agree outside a ball of radius h around x_0 but differ materially at x_0.

3 The lower bound

Theorem 2 (Honest local coverage forces inflation)**.** *Fix $0 < \alpha < 1$ and $0 \le \eta < \frac{1}{2}$, and let Assumptions 1–3 hold. Then there is a constant $c > 0$, depending only on the model constants $(\alpha, \eta, K, c_q, L, g_-, g_+, \psi)$, such that every T_n satisfying honest local coverage (2) over $\mathcal{H}^s(L)$ obeys*

$$\sup_{\theta \in \mathcal{H}^s(L)} S_\theta(T_n) \;\ge\; c\,\big(n\,g(x_0)\big)^{-s/(2s+d)}. \tag{4}$$

No finite-sample honestly-local procedure is oracle-sharp uniformly over the class; its expected excess threshold is at least the local nonparametric estimation scale.

Proof. Take the constant baseline $\theta_0 \equiv u_0$ and the bump $\theta_1(x) = u_0 + \delta\,\psi((x - x_0)/h)$ of Assumption 3, with the sign of $\psi(0)$ chosen so that $q_\alpha(\theta_1(x_0)) > q_\alpha(\theta_0(x_0))$. Both lie in $\mathcal{H}^s(L)$.

Indistinguishability. The two laws differ only on the bump support, of covariate measure $\Theta(h^d)$, carrying $\Theta(n\,g(x_0)\,h^d)$ calibration points in expectation by Assumption 1. Each contributes at most $K\delta^2$ to the Kullback–Leibler divergence by Assumption 2, so for product measures over the calibration sample

$$\mathrm{KL}\big(P^n_{\theta_1}, P^n_{\theta_0}\big) \;\lesssim\; n\,g(x_0)\,h^d\,\delta^2.$$

Choose $h \asymp \big(ng(x_0)\big)^{-1/(2s+d)}$, hence $\delta \asymp L\,h^s \asymp \big(ng(x_0)\big)^{-s/(2s+d)}$. Then $ng(x_0)h^d\delta^2 = O(1)$, so by Pinsker the total-variation distance is bounded, $\mathrm{TV}(P^n_{\theta_1}, P^n_{\theta_0}) \le \tau$ for a constant $\tau < 1-\eta$ (shrink the bump amplitude by a fixed factor if needed to make τ that small).

Transfer. Honest validity (2) under θ_1 gives $\mathbb{P}_{\theta_1}\{T_n(x_0) \ge q_\alpha(\theta_1(x_0))\} \ge 1-\eta$. The event $\{T_n(x_0) \ge q_\alpha(\theta_1(x_0))\}$ is a function of the calibration sample, so its probability changes by at most TV between the two laws:

$$\mathbb{P}_{\theta_0}\big\{T_n(x_0) \ge q_\alpha(\theta_1(x_0))\big\} \;\ge\; (1-\eta) - \tau \;>\; 0.$$

Inflation. Under the baseline the oracle threshold is $q_\alpha(\theta_0(x_0))$, so

$$\begin{aligned}
S_{\theta_0}(T_n) &= \mathbb{E}_{\theta_0}\big[(T_n(x_0) - q_\alpha(\theta_0(x_0)))_+\big] \\
&\ge \big[q_\alpha(\theta_1(x_0)) - q_\alpha(\theta_0(x_0))\big]\cdot \mathbb{P}_{\theta_0}\big\{T_n(x_0) \ge q_\alpha(\theta_1(x_0))\big\} \\
&\ge c_q\,\delta\,\big((1-\eta)-\tau\big) \;\gtrsim\; \big(ng(x_0)\big)^{-s/(2s+d)},
\end{aligned}$$

using the lower quantile-sensitivity bound of Assumption 2 for $q_\alpha(\theta_1(x_0)) - q_\alpha(\theta_0(x_0)) \ge c_q\,\delta\,|\psi(0)|$. Since $\theta_0 \in \mathcal{H}^s(L)$, the supremum in (4) is at least this, proving the claim. $\square$

The mechanism is exactly the intended one. The data cannot tell a smooth baseline from a small, local increase in residual uncertainty. Exact local protection against the increase therefore forces the set to stay inflated even when the baseline is true, and the size of the forced inflation is the amplitude of the largest undetectable bump.

4 Attainability, so the bound is not vacuous

The lower bound is only interesting if an honest procedure exists whose inflation matches it, so that Theorem 2 describes a real optimum rather than an empty constraint. One does, by the standard bias-plus-margin construction. Estimate the local oracle threshold by a boxcar (or kernel) quantile $\widehat{q}(x_0)$ over the ball $B(x_0, h)$, holding $m_h \asymp n\,g(x_0)h^d$ effective calibration points. Under Assumptions 1–3 its bias is at most C_qLh^s (quantile sensitivity applied to the Hölder variation of θ over the ball) and its stochastic fluctuation is $O_P(m_h^{-1/2})$. Set

$$T_n(x_0) \;=\; \widehat{q}(x_0) \;+\; C_qLh^s \;+\; \kappa\, m_h^{-1/2},$$

with $\kappa = \kappa(\eta)$ a fixed multiple of the fluctuation scale. The deterministic bias term covers the systematic error and the margin covers the stochastic error with probability $1-\eta$, so T_n is honestly locally valid (2), and its excess threshold is $O\big(Lh^s + m_h^{-1/2}\big)$. Minimizing over h gives the modulus (5), so the excess matches (4) up to constants. The honest optimum is therefore exactly the local modulus, achieved by inflating a local quantile estimate by its own bias and margin, and Theorem 2 says no honest procedure does better.

5 The bandwidth reading

The rate is the local modulus of continuity,

$$\big(ng(x_0)\big)^{-s/(2s+d)} \;\asymp\; \inf_{h>0}\Big\{\,L\,h^s \;+\; \big(n\,g(x_0)\,h^d\big)^{-1/2}\Big\}, \tag{5}$$

the sum of the largest undetectable local heterogeneity, Lh^s, and the local sampling scale, $(ng(x_0)h^d)^{-1/2}$. Writing $m_h \asymp ng(x_0)h^d$ for the effective local sample size, the honesty cost is $Lh^s + m_h^{-1/2}$: as data thin, both the best neighbourhood and the unavoidable inflation grow. This is the small-sample statement, with no appeal to uniform pooling. Exact local validity does not force wide neighbourhoods; it forces protection against locally undetectable alternatives, whose width is the modulus.

6 Unknown smoothness: honesty cannot adapt

Over a single known class the lower bound (4) is attainable, so coverage and minimax-rate sharpness are not incompatible there. The genuine separation is adaptive.

Theorem 3 (No adaptation under honesty)**.** *Let $0 < s_0 < s_1$ and require honest local coverage over $\mathcal{H}^{s_0}(L) \cup \mathcal{H}^{s_1}(L)$. Then there is $c > 0$ with*

$$\sup_{\theta\in\mathcal{H}^{s_1}(L)} S_\theta(T_n) \;\geq\; c\,\big(ng(x_0)\big)^{-s_0/(2s_0+d)}, \tag{6}$$

which is strictly slower than the smooth-class rate $\big(ng(x_0)\big)^{-s_1/(2s_1+d)}$. An honest set cannot achieve the faster sharpness available on the smoother subclass.

Proof. Take the baseline $\theta_0 \equiv u_0$, which is constant and hence lies in $\mathcal{H}^{s_1}(L)$, and the indistinguishable bump alternative from the rougher class $\mathcal{H}^{s_0}(L)$ at $h \asymp (ng(x_0))^{-1/(2s_0+d)}$, $\delta \asymp (ng(x_0))^{-s_0/(2s_0+d)}$. The proof of Theorem 2 applies verbatim and forces $S_{\theta_0}(T_n) \gtrsim \delta \asymp (ng(x_0))^{-s_0/(2s_0+d)}$, even though $\theta_0 \in \mathcal{H}^{s_1}(L)$. □

This is the honesty–adaptation obstruction familiar from nonparametric confidence bands (Low, 1997; Cai and Low, 2004). A sharp point estimator can read apparent smoothness off the data and choose a favourable bandwidth. An honest coverage guarantee cannot trust apparent smoothness, because a rough local alternative is statistically indistinguishable, so it must carry the rougher width. This is the strongest form of the thesis: the local estimator is sharper not by a better estimate but by declining to insure against what it cannot see.

Remark 4 (Average validity is weaker)*.* If validity means only $\inf_\theta \mathbb{E}_\theta F_{\theta(x_0)}(T_n) \geq 1-\alpha$, averaging over calibration samples, a procedure may compensate a too-narrow set on some samples with a wide set on others, and the first-order argument weakens. For a family in which $t \mapsto F_u(t)$ is concave over the range of T_n and uniformly strictly concave near q_α (equivalently, T_n concentrates near q_α with high probability), Jensen gives $\mathbb{E}_\theta T_n \geq q_\alpha(\theta(x_0))$ and a curvature bound $\mathrm{Var}(T_n) \lesssim \mathbb{E}_\theta T_n - q_\alpha(\theta(x_0))$, so small expected excess would make T_n a low-MSE estimator of the conditional quantile; the minimax MSE bound then yields $\sup_\theta[\mathbb{E}_\theta T_n - q_\alpha(\theta(x_0))] \gtrsim (ng(x_0))^{-2s/(2s+d)}$. The squared exponent is not an accident: average coverage is materially weaker than honest per-sample coverage.

7 What it says for conformal prediction

The obstruction is not exchangeability, weighting, or pooling; localized conformal sets that reweight by proximity are marginally valid and can be sharp. The obstruction is indistinguishability under an honesty requirement. Four consequences follow.

Exact marginal conformal validity coexists with local smoothing and asymptotic efficiency. Exact local honest validity necessarily inflates the set by at least the local modulus $(ng(x_0))^{-s/(2s+d)}$. A plug-in local residual estimate can be sharper because it does not protect against every statistically indistinguishable local alternative, only against what it can see. And when smoothness is unknown, honesty forbids adaptation to the faster rate a smoother law would allow.

The applied reading is the one experience already supplies. At small effective local sample size m_h, the modulus $Lh^s + m_h^{-1/2}$ is large, the honest set is inflated by a correspondingly large amount, and a kernel or local-likelihood estimate of the residual law, which forgoes honest local coverage for asymptotic validity, is the better forecast. The honest guarantee is not free and it is not distribution-free at a point; its price is the sharpness it must surrender to insure against the alternatives the data cannot rule out.

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